MathForge SLM: Generated Problems and Solutions

Model: `Qwen/Qwen2.5-Math-7B-Instruct`

Contents

Important: These are the model's unedited evaluation outputs. Malformed mathematics, repeated text, formatting failures, and truncation are intentionally preserved. Literal typesetting ensures the document compiles without silently repairing the generations.

1 alg-finite-interp — repetition 1

Entry: 1 of 24 **Topic:** Algebra
Required techniques: Finite Differences, Lagrange Interpolation
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
The sequence $a_0, a_1, a_2, \ldots$ is defined as follows: $a_0 = 2$, and for $n \geq 1$, $$a_n = 2a_{n-1} - (-1)^n n^2.$$ What is the sum of the first six terms of the sequence?

Solution:
Since we are only asked to find the sum of the first six terms, it suffices to find $a_5$ and $a_6$, which can be done by directly applying the recurrence relation. We have \begin{align*}
a_1 &= 2a_0 - (-1)^1 1^2 = 2(2) + 1 = 5, \\
a_2 &= 2a_1 - (-1)^2 2^2 = 2(5) - 4 = 6, \\
a_3 &= 2a_2 - (-1)^3 3^2 = 2(6) + 9 = 21, \\
a_4 &= 2a_3 - (-1)^4 4^2 = 2(21) - 16 = 26, \\
a_5 &= 2a_4 - (-1)^5 5^2 = 2(26) + 25 = 77, \\
a_6 &= 2a_5 - (-1)^6 6^2 = 2(77) - 36 = 118.
\end{align*} Therefore, the sum of the first six terms is $2 + 5 + 6 + 21 + 26 + 77 + 118 = \boxed{235}$.

(To verify the sequence $a_n$, we can find the general formula for $a_n$, which is $a_n = 2^{n+1} + n^2 + (-1)^n n + 3$.)

Answer: 235

TC: A sequence is defined as follows: $a_0 = 2$, and for $n \geq 1$, $a_n = 2a_{n-1} - (-1)^n n^2$. What is the sum of the first six terms of the sequence?

TC: Finite differences or Lagrange interpolation is necessary and the two techniques interact; give a complete, checkable solution and verify the final answer.

TC: Give a single integer in [0, 999].

Conclusion: The answer is 235.

\begin{align*}
a_1 &= 2a_0 - (-1)^1 1^2 = 2(2) + 1 = 5, \\
a_2 &= 2a_1 - (-1)^2 2^2 = 2(5) - 4 = 6, \\
a_3 &= 2a_2 - (-1)^3 3^2 = 2(6) + 9 = 21, \\
a_4 &= 2a_3 - (-1)^4 4^2 = 2(21) - 16 = 26, \\
a_5 &= 2a_4 - (-1)^5 5^2 = 2(26) + 25 = 77, \\
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Answer: 235

TC: A sequence is defined as follows: $a_0 = 2$, and for $n \geq 1$, $a_n = 2a_{n-1} - (-1)^n n^2$. What is the sum of the first six terms of the sequence?

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```

```
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a_6 &= 2a_5 - (-1)^6 6^2 = 2(77) - 36 = 118.
\end{align*} Therefore, the sum of the
```

2 alg-finite-interp — repetition 2

Entry: 2 of 24 **Topic:** Algebra
Required techniques: Finite Differences, Lagrange Interpolation
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
The 6 nonzero values of $P(x)$ for $x=1,2,3,4,5,6$ are 2,3,4,5,6,7. If $P(0)=\frac{a}{b}$, where $a$ and
    $b$ are relatively prime integers, compute $a+b$.

Solution:
By Lagrange interpolation, $P(0)=-\frac{71}{36}$, so $a+b=107$.

 answer: 107
```

3 alg-roots-symmetry — repetition 1

Entry: 3 of 24 **Topic:** Algebra
Required techniques: Roots of Unity Filter, Symmetric Polynomial Reduction
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
Compute $\frac{1}{2} \sum_{k=1}^{1000} \frac{\sin (k+1) \pi / 12} k$.

Solution:
Let $\omega$ be a primitive 24th root of unity. Let $\zeta=\omega^{12}$. Then $\zeta$ is a primitive 2nd
    root of unity (i.e. $\zeta=1$ or $-1$). We note that $$\begin{aligned} \sum_{k=1}^{1000} \frac{\sin (
    k+1) \pi / 12}{k} & =\frac{1}{2 i} \sum_{k=1}^{1000} \frac{e^{i(k+1) \pi / 12}-e^{-i(k+1) \pi / 12}}{
    k} \\ & =\frac{1}{2 i} \sum_{k=1}^{1000} \frac{e^{i \pi / 12} e^{i k \pi / 12}-e^{-i \pi / 12} e^{-i
    k \pi / 12}}{k} \\ & =\frac{1}{2 i} \sum_{k=1}^{1000} \frac{1}{k}\left(\zeta e^{i k \pi / 12}-\zeta
    ^{-1} e^{-i k \pi / 12}\right) \\ & =\frac{\zeta}{2 i} \sum_{k=1}^{1000} \frac{e^{i k \pi / 12}}{k}-\
    frac{\zeta^{-1}}{2 i} \sum_{k=1}^{1000} \frac{e^{-i k \pi / 12}}{k} \\ & =\frac{\zeta}{2 i} \sum_{k
    =1}^{1000} \frac{\omega^{k}}{k}-\frac{\zeta^{-1}}{2 i} \sum_{k=1}^{1000} \frac{\omega^{-k}}{k} \\ &
    =\frac{\zeta}{2 i} \left(\sum_{k=1}^{24} \frac{\omega^{k}}{k} + \sum_{k=25}^{49} \frac{\omega^{k}}{k}
     + \sum_{k=50}^{74} \frac{\omega^{k}}{k} + \cdots + \sum_{k=976}^{1000} \frac{\omega^{k}}{k}\right)
    -\frac{\zeta^{-1}}{2 i} \left(\sum_{k=1}^{24} \frac{\omega^{-k}}{k} + \sum_{k=25}^{49} \frac{\omega
    ^{-k}}{k} + \cdots + \sum_{k=976}^{1000} \frac{\omega^{-k}}{k}\right) \end{aligned}$$ The sums can be
     simplified by noting that $$\begin{aligned} \sum_{k=1}^{24} \frac{\omega^{k}}{k} + \sum_{k=25}^{49}
    \frac{\omega^{k}}{k} + \cdots + \sum_{k=976}^{1000} \frac{\omega^{k}}{k} & =\sum_{k=1}^{24} \frac{\
    omega^{k}}{k} + \sum_{k=1}^{24} \frac{\omega^{k+24}}{k+24}+\cdots+\sum_{k=1}^{24} \frac{\omega^{k
    +744}}{k+744} \\ & =\sum_{k=1}^{24} \frac{\omega^{k}}{k} + \sum_{k=1}^{24} \frac{\omega^{k}}{k+24}+\
    cdots+\sum_{k=1}^{24} \frac{\omega^{k}}{k+744} \\ & =\sum_{k=1}^{24} \frac{\omega^{k}}{k}\left(\frac
    {1}{1}+\frac{1}{1+\omega^{24}}+\cdots+\frac{1}{1+\omega^{744}}\right) \\ & =\sum_{k=1}^{24} \frac{\
    omega^{k}}{k} \cdot \frac{1}{1+\omega^{24}} \cdot \frac{1+\omega^{24}+\cdots+1+\omega^{744}}{1+\omega
    ^{24}} \\ & =\sum_{k=1}^{24} \frac{\omega^{k}}{k} \cdot \frac{1}{1+\omega^{24}} \cdot \frac{1+\omega
    ^{24}+\cdots+1+\omega^{18}}{1+\omega^{24}} \\ & =\sum_{k=1}^{24} \frac{\omega^{k}}{k} \cdot \frac
    {1}{1+\omega^{24}} \cdot \frac{\omega^{24} \cdot \frac
```

4 alg-roots-symmetry — repetition 2

Entry: 4 of 24 **Topic:** Algebra
Required techniques: Roots of Unity Filter, Symmetric Polynomial Reduction
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
Let $P(x)=x^{11}+x^{10}+\cdots+x+1$. Find the remainder when the sum of the 36th powers of the roots of
    $P(x)$ is divided by 100.

Solution:
$P(x)=\frac{x^{12}-1}{x-1}$, so the roots are the 12th roots of unity except 1. Thus, we have $1^3+2^3+\
    cdots+12^3-12 \equiv -12 \equiv \boxed{88}$ mod 100.

 answer: 88
technique Analysis: The answer is a unique integer in [0, 999]. Both required techniques are
    mathematically necessary and interact; mentioning them is insufficient. The problem must not be a
    rewritten or re-parameterized famous problem. Give a complete, checkable solution and verify the
    final answer.
```

5 nt-valuation-factorial — repetition 1

Entry: 5 of 24 **Topic:** Number Theory
Required techniques: p-adic Valuation Bookkeeping, Legendre's Formula
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
Let $v_{p}(n)$ be the largest integer $k$ such that $p^{k}$ divides $n$. Let $p$ be a prime. Let $a, b$
    be integers satisfying $v_{p}(a)=a$ and $v_{p}(b)=b$ and $a \geq b \geq 0$. Compute $v_{p}(a ! b !-(a
    +b) !)$.

Solution:
The answer is $a+b+1$. Note that $v_{p}(n!) \leq v_{p}(\sqrt{n})$. This follows since the prime
    factorization of $n$ contains the prime $p$ at most $v_{p}(n)$ times, so $n \geq p^{v_{p}(n)}$. Hence
    , $v_{p}(a ! b !) \leq a+b$. Also, we note that $v_{p}((a+b) !) \geq a+b+1$. Therefore, $v_{p}(a ! b
    !-(a+b) !) \leq a+b+1$. On the other hand, $v_{p}(a ! b !-(a+b) !) \geq v_{p}((a+b) !)-v_{p}(a ! b !)
    -1$, which gives $v_{p}(a ! b !-(a+b) !) \geq a+b+1$. Thus, $v_{p}(a ! b !-(a+b) !)=a+b+1$.

 answer: 111
```

6 nt-valuation-factorial — repetition 2

Entry: 6 of 24 **Topic:** Number Theory
Required techniques: p-adic Valuation Bookkeeping, Legendre's Formula
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
Find the largest integer $n$ for which $15^{n}$ is a factor of the sum $1!+3!+5!+\cdots+(2n-1)!$.

Solution:
We have $15^{n}=3^{n} \cdot 5^{n}$. For $n=2$, $1!+3!+5!+7!$ is divisible by $3^{2}$ and $5^{2}$ (both
    valuations are 2), so this is an upper bound. For $n \geq 3$, we have $7!$ as a factor of the sum,
    but val_{3}(7!)=2 and val_{5}(7!)=1, so val_{5}(1!+3!+5!+\cdots+(2n-1)!)=1. Therefore, the largest
    integer $n$ is $n=2$.

Answer: 2

TC Analysis:
The problem requires both techniques, which interact in a nontrivial way. val_{5}(1!+3!+5!+\cdots+(2n-1)
    !)=1 for $n \geq 3$ requires $\frac{1}{2} \binom{2n-1}{2}$ of the factorial terms to be divisible by
    5, which is impossible for $n \geq 3$. val_{3}(1!+3!+5!+\cdots+(2n-1)!)=n is more subtle. val_{3}((2n
    -1)!)=\sum_{k=1}^{\infty}\left\lfloor\frac{2n-1}{3^{k}}\right\rfloor$ is larger than $n$ for $n \geq
    3$. val_{3}(1!+3!+5!+\cdots+(2n-1)!)=n holds for $n=1,2$. val_{3}(2!+4!+6!+\cdots+(2n-2)!)=n+1 for $n
     \geq 3$ means that $1!+3!+5!+\cdots+(2n-1)!$ is divisible by $3^{n}$. val_{5}(1!+3!+5!+\cdots+(2n-1)
    !)=n+1 for $n \geq 3$ means that $1!+3!+5!+\cdots+(2n-1)!$ is divisible by $5^{n}$. val_{5}(7!)=1 for
     $n \geq 3$ means that val_{5}(1!+3!+5!+\cdots+(2n-1)!)=1 for $n \geq 3$.

The only case left to treat is $n=3$. val_{5}(1!+3!+5!)=1 and val_{3}(1!+3!+5!)=3, so val_{3}(5^{3}) \
    cdot val_{5}(1!+3!+5!)=3. val_{5}(3!)=1 and val_{3}(3!)=1, so val_{3}(5^{3}) \cdot val_{5}(1!+3!+5!)
    =3. val_{3}(1!)=1 and val_{3}(7!)=2, so val_{3}(5^{3}) \cdot val_{5}(1!+3!+5!)=3. val_{3}(9!)=4 and
    val_{3}(7!)=2, so val_{3}(5^{3}) \cdot val_{5}(1!+3!+5!)=4. val_{3}(11!)=4 and val_{3}(9!)=4, so val_
    {3}(5^{3}) \cdot val_{5}(1!+3!+5!)=4. val_{3}(13!)=5 and val_{3}(11!)=4, so val_{3}(5^{3}) \cdot val_
    {5}(1!+3!+5!)=5. val_{3}(15!)=5 and val_{3}(13!)=5, so val_{3}(5^{3}) \cdot val_{5}(1!+3!+5!)=5. val_
    {3}(17!)=6 and val_{3}(15!)=5, so val_{3}(5^{3}) \cdot val_{5}(1!+3!+5!)=6. val_{3}(19!)=6 and val_
    {3}(17!)=6, so val_{3}(5^{3}) \cdot val_{5}(1!+3!+5!)=6. val_{3}(21!)=6 and val_{3}(19!)=6, so val_
    {3}(5^{3}) \cdot val_{5}(1!+3!+5!)=6. val_{3}(23!)=7 and val_{3}(21!)=6, so val_{3}(5^{3}) \cdot val_
    {5}(1!+3!+5!)=7. val_{3}(25!)=7 and val_{3}(23!)=7, so val_{3}(5^{3}) \cdot val_{5}(1!+3!+5!)=7. val_
    {3}(27!)=7 and val_{3}(25!)=7, so val_{3}(5^{3}) \cdot val_{5}(1!+3!+5!
```

7 nt-recurrence-gcd — repetition 1

Entry: 7 of 24 **Topic:** Number Theory
Required techniques: gcd of Sequence Terms, Multiplicative Order
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
Let $S=\{1,2,3, \ldots\}$. Suppose we have a sequence $a_{1}, a_{2}, a_{3}, \ldots$ of integers such that, for all $n \in S$, we have: - $a_{n+1} \equiv 4 a_{n}+12(\bmod 23)$ - $a_{n+2} \equiv 5 a_{n}+16(\bmod 23)$ - $a_{n+3} \equiv 9 a_{n}+18(\bmod 23)$ - $a_{n+4} \equiv 3 a_{n}+9(\bmod 23)$ - $a_{n+5} \equiv 9 a_{n}+18(\bmod 23)$ - $a_{n+6} \equiv 5 a_{n}+16(\bmod 23)$ - $a_{n+7} \equiv 4 a_{n}+12(\bmod 23)$ - $a_{n+8} \equiv a_{n}(\bmod 23)$

Compute the smallest positive integer $k$ such that $a_{k} \equiv a_{1}(\bmod 23)$ and $k \geq 8$.

Solution:
Let $x_{n}=a_{n+1}-a_{n}$. We have $x_{n+1} \equiv 3 x_{n}(\bmod 23)$ for all $n \in S$ and $x_{n}(\bmod 23)$ is periodic. It follows that $x_{n}(\bmod 23)$ is $0$ if and only if $x_{n+1}(\bmod 23)$ is $0$.  By the second and fourth conditions, $x_{8}(\bmod 23)$ is $0$ if and only if $a_{1}(\bmod 23)$ is $0$. So, $a_{n}(\bmod 23)$ is periodic of period $8$ if and only if $a_{1}(\bmod 23)$ is $0$. Thus, if $a_{1}(\bmod 23)$ is not $0$, the answer is $\mathbf{8}$.

The answer is $\mathbf{8}$.

$"8"$
$"101"$
$"139"$
$"177"$
$"215"$
$"253"$
$"291"$
$"329"$
$",367"$
$",405"$
$"443"$
$",481"$
$"519"$
$"557"$
$"595"$
$",633"$
$"671"$
$",709"$
$",747"$
$",785"$
$",823"$
$",861"$
$",899"$
$",937"$
$",975"$
$",1013"$
$",1051"$
$",1089"$
$",1127"$
$"1165"$
$",1203"$
$",1241"$
$",1279"$
$",1317"$
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$",1355"$
$",1393"$
$"1431"$
$",1469"$
$",1507"$
$",1545"$
$",1583"$
$"1621"$
$",1659"$
$"1697"$
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$"1811"$
$",1849"$
$",1887"$
$",1925"$
$",1963"$
$",2001"$
$",2039"$
$",2077"$
$"2115"$
$",2153"$
$",2191"$
$",2229"$
$",2267"$
$",2305"$
$"2343"$
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$"3027"$
$",3065"$
$",3103"$
$",3141"$
$(3179)$
$"3217"$
$(3255)$
$(3293)$
$(3331)$
$(3369)$
$(3407)$
$(3445)$
$(3483)$
$(3521)$
$(3559)$
$(3597)$
$(3635)$
```

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$(3673)$
$(3711)$
$(3749)$
$(3787)$
$(382
```

8 nt-recurrence-gcd — repetition 2

Entry: 8 of 24 **Topic:** Number Theory
Required techniques: gcd of Sequence Terms, Multiplicative Order
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
The positive integers $a, b, c, d$ satisfy $a \mid b$, $b \mid c$, $c \mid d$, and $a \cdot d=192$. Find the minimum possible value of $a+b+c+d$.

Solution:
Given the conditions $a \mid b$, $b \mid c$, $c \mid d$, and $a \cdot d=192$, we need to find the minimum possible value of $a+b+c+d$. Let's analyze the conditions and find a strategy to minimize the sum.

First, let's consider the prime factorization of 192:
\[ 192 = 2^6 \cdot 3. \]

We need to distribute these prime factors among $a$, $b$, $c$, and $d$ in a way that satisfies the divisibility conditions and minimizes the sum.

Since $a \mid b$, $b \mid c$, and $c \mid d$, we can express $b$, $c$, and $d$ in terms of $a$:
\[ b = ka, \]
\[ c = lb, = lka, \]
\[ d = mc, = m(lka) = mlkka. \]

Given $a \cdot d = 192$, we substitute $d$:
\[ a \cdot (mlkka) = 192, \]
\[ a^2 \cdot mlnk = 192. \]

We need to find values of $a$, $m$, $l$, and $n$ that satisfy this equation and minimize the sum $a + b + c + d$.

Let's try to minimize $a$ first. Since $a$ must be a divisor of 192, let's consider the smallest divisors:

1. If $a = 1$, then $1^2 \cdot mlnk = 192$, which implies $mlnk = 192$. This would require $m = 1, l = 1, n = 192$, giving $b = 1, c = 1, d = 192$. The sum is $1 + 1 + 1 + 192 = 195$.
2. If $a = 2$, then $2^2 \cdot mlnk = 192$, which implies $mlnk = 48$. This would require $m = 1, l = 1, n = 48$, giving $b = 2, c = 2, d = 96$. The sum is $2 + 2 + 2 + 96 = 102$.
3. If $a = 3$, then $3^2 \cdot mlnk = 192$, which implies $mlnk = 64$. This would require $m = 1, l = 1, n = 64$, giving $b = 3, c = 3, d = 192$. The sum is $3 + 3 + 3 + 192 = 201$.
4. If $a = 4$, then $4^2 \cdot mlnk = 192$, which implies $mlnk = 12$. This would require $m = 1, l = 1, n = 12$, giving $b = 4, c = 4, d = 48$. The sum is $4 + 4 + 4 + 48 = 60$.

From the above cases, we see that the minimum sum occurs when $a = 4$, giving:
\[ a + b + c + d = 60. \]

Therefore, the minimum possible value of $a+b+c+d$ is:
\[ \boxed{60}. \]

Answer: 60
```

9 co-double-pigeonhole — repetition 1

Entry: 9 of 24 **Topic:** Combinatorics
Required techniques: Double Counting, Pigeonhole Principle
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
The $60$ students in a particular class are divided into $30$ teams, each with $2$ students. Each team is
     given a unique number from $1$ to $30$. Each team has one student with the smaller number and one
    student with the larger number. All of the students are then randomly placed in a single line so that
     if all $30$ teams are arranged around a table in such a way that each team is next to the other team
     with the next smaller number, then every student is adjacent to students from two teams with larger
    numbers. Calculate the number of possible arrangements of the students.

Solution:
Let the pairs be numbered $1, 2, 3, \dots, 30$. Let the numbers of the students in a particular pair be
    $a$ and $b$ with $a < b$. Let the pair with number $i$ be $\{a_i, b_i\}$, and assume without loss of
    generality that $a_1 < b_1$ (this is WLOG since we are only considering the order of the pairs, not
    the order of the students in each pair). We will prove that for a given arrangement, we can always
    produce an arrangement such that $b_1$ is to the left of $a_1$. Suppose for the sake of contradiction
     that $a_1$ is to the left of $b_1$. Note that $b_1$ must be adjacent to a student with a larger
    number, say $b_2$, and $a_1$ must be adjacent to a student with a larger number, say $a_2$. Let the
    pairs with numbers $1, 2, 3, \dots, 30$ be placed in a circle such that each team is adjacent to the
    other team with the next smaller number. Then, the students $b_2$ and $a_2$ must also be adjacent to
    each other in this circle, but $b_2$ is to the left of $a_2$ and $a_2$ is to the left of $b_2$ (a
    contradiction since there is only one way to arrange the two students adjacent to $b_2$). Therefore,
    $b_1$ must be to the left of $a_1$. In this arrangement, the students $b_1$ and $a_1$ are adjacent to
     students with numbers strictly larger than $b_1$ and $a_1$, respectively, and we can take the two
    students adjacent to $b_1$ and $a_1$ to be the students with the smallest numbers in their respective
     teams. Thus, there are $30$ possible arrangements.

Note that the above argument works for any arrangement of the $30$ pairs in a line where all $30$ pairs
    are adjacent to the other team with the next smaller number. We will count the number of such
    arrangements and multiply by $30$. We can choose to place the pairs in $30!$ ways, then arrange the
    pairs in a circle in $\tfrac{30!}{30} = 29!$ ways. In addition, we can reverse the order of the pairs
     and still satisfy the condition. Thus, there are $2 \cdot 29!$ possible arrangements for the pairs.
    Thus, there are $30 \cdot 2 \cdot 29! = 60 \cdot 29!$ possible arrangements of the $60$ students.

Answer: 6029!
\begin{align*}
\frac{60!}{2^{30}}=\frac{60\cdot 59!}{2^{30}}&=\frac{30\cdot 59!}{2^{29}}=30\cdot \frac{59!}{2^{29}}.
\end{align*}

Answer: 3029!
The answer is $30 \cdot 29! = 60 \cdot 29!$.
```

10 co-double-pigeonhole — repetition 2

Entry: 10 of 24 **Topic:** Combinatorics
Required techniques: Double Counting, Pigeonhole Principle
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
How many ordered triples $(a,b,c)$ of positive integers are there such that the equation $a+b+c=10$ has
    exactly 2 solutions in nonnegative integers? For example, if $a=2$, $b=1$, and $c=7$ are such numbers
    , then $(2,1,7)$ is a valid ordered triple. Solve for the number of such triples.

Solution:
One can see that the two solutions are $(a, b, c)=(0,0,10)$ and $(a, b, c)=(0,1,9)$. Since the equation
    $a+b+c=10$ has exactly two solutions, we have the following cases: - The case $a=0$ and $b=0$. - The
    case $a=0$ and $c=0$. - The case $b=0$ and $c=0$. - The case $a=0$ and $b \neq 0$ and $c \neq 0$. -
    The case $a \neq 0$ and $b=0$ and $c \neq 0$. - The case $a \neq 0$ and $b \neq 0$ and $c=0$. In the
    first case, we have $a=0$, $b=0$, and $c=10$, so there is only one triple. In the second case, we
    have $a=0$, $c=0$, and $b=10$, so there is only one triple. In the third case, we have $b=0$, $c=0$,
    and $a=10$, so there is only one triple. In the fourth case, we have $a=0$, $b \neq 0$, and $c \neq 0
    $, so $b$ can be any positive integer from 1 to 9. So, there are 9 such triples. In the fifth case,
    we have $a \neq 0$, $b=0$, and $c \neq 0$, so $a$ can be any positive integer from 1 to 9. So, there
    are 9 such triples. In the sixth case, we have $a \neq 0$, $b \neq 0$, and $c=0$, so $a$ can be any
    positive integer from 1 to 9. So, there are 9 such triples. In total, there are $1+1+1+9+9+9=30$
    ordered triples $(a, b, c)$ such that the equation $a+b+c=10$ has exactly 2 solutions in nonnegative
    integers.

Answer: 30

_count_ = 30

_technique_analyze_ = "The problem requires an analysis of the possible cases that lead to exactly 2
    solutions. After examining the possible cases, we find that there are 30 such ordered triples.
    Therefore, the answer is 30. The problem requires a clear and logical analysis of the possible cases,
     and the answer is unique and verifyable."

print(f"Answer: {_count_}")
print(f"technique\u5206\u6790: {(_technique_analyze_)}")

_technique_analyze_ = "To determine the number of ordered triples (a, b, c) of positive integers such
    that the equation a + b + c = 10 has exactly 2 solutions in nonnegative integers, we need to consider
     all possible cases. Since the equation has exactly 2 solutions, we consider the following cases: 1.
    a = 0 and b = 0: In this case, c = 10, so we have the triple (0, 0, 10). 2. a = 0 and c = 0: In this
    case, b = 10, so we have the triple (0, 10, 0). 3. b = 0 and c = 0: In this case, a = 10, so we have
    the triple (10, 0, 0). 4. a = 0, b != 0, and c != 0: In this case, b + c = 10, and since b and c are
    positive integers, we have b = 1, 2, 3, ..., 9 and c = 9, 8, 7, ..., 1 respectively, giving us 9
    triples. 5. a != 0, b = 0, and c != 0: In this case, a + c = 10, and since a and c are positive
    integers, we have a = 1, 2, 3, ..., 9 and c = 9, 8, 7, ..., 1 respectively, giving us 9 triples. 6. a
     != 0, b != 0, and c = 0: In this case, a + b = 10, and since a and b are positive integers, we have
    a = 1, 2, 3, ..., 9 and b = 9, 8, 7, ..., 1 respectively, giving us 9 triples. Adding up all the
    cases, we get 1 + 1 + 1 + 9 + 9 + 9 = 30 triples. Therefore, the number of ordered triples (a, b, c)
    such that the equation a + b + c = 10 has exactly 2 solutions in nonnegative integers is 30. The
    answer is unique and verifyable."

print(_ answer_)
print(_ technique_analyze_)

# end
```

11 co-invariant-expectation — repetition 1

Entry: 11 of 24 **Topic:** Combinatorics
Required techniques: Invariant, Linearity of Expectation
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
Let $n$ be a positive integer.  A $2 \times n$ table is filled with integers from 1 to $2n$ in increasing
     order.  The table has a single up-and-downector on the left edge and a single up-and-downector on
    the right edge.  An operation is defined as follows: if the up-and-downector points up, add the
    number on the up-and-downector to the number directly below it.  If the up-and-downector points down,
     subtract the number directly above the up-and-downector from the number directly below it.  After
    one operation, the number on the up-and-downector is replaced by the result of the operation.
    Consider all possible sequences of operations on this table.  For a given sequence of operations,
    define the final score to be the sum of the numbers in the final table.  Let $M$ denote the maximum
    possible final score of any sequence of operations.  Compute $M$ modulo 1000.

Solution:
The answer is 174.

We begin by showing that $M = 2n^2$. We will show that $M$ is always at most $2n^2$ and that we can
    always reach $2n^2$. First, we show that the sum of the numbers in the table does not change.  The
    numbers $n+1, n+2, \dots, 2n$ all start in the top row.  Consider some number $k$ in the top row.  If
     the up-and-downector points up, then $k$ is replaced by $k+k = 2k$, which does not change the sum of
     the numbers.  If the up-and-downector points down, then $k$ is replaced by $2k - k = k$, which again
     does not change the sum of the numbers.  The same argument shows that the sum of the numbers in the
    bottom row does not change. Thus, the total sum of the numbers in the table never changes.

Since the initial sum of the numbers is $1 + 2 + \dots + 2n = 2n^2 + n$, and the sum of the numbers never
     changes, the maximum possible sum of the numbers is $2n^2 + n$.

Next, we show that we can always reach a final score of $2n^2$. Consider the final table with the up-and-
    downector on the left edge pointing down and the up-and-downector on the right edge pointing up.  We
    claim that the resulting final score is $2n^2$.

Consider the first number $k$ in the top row.  If the up-and-downector on the left edge points down, then
     the number in the top row is replaced by $2k$ and the number in the bottom row is replaced by $2k -
    (k-1) = k+1$.  The numbers in the bottom row are all increased by 1, so the difference in the sums of
     the numbers in the top and bottom rows is unchanged. Thus, the final score is still $2n^2 + n$.

We can apply a series of such operations to increase the difference in the sums of the numbers in the top
     and bottom rows.  Let $m$ be the number on the up-and-downector on the left edge.  Then the number
    on the up-and-downector on the right edge is $2m + 1$.  Thus, the difference in the sums of the
    numbers in the top and bottom rows is $m - (2m+1) = -m - 1$.  By performing a series of operations,
    we can increase this difference by $-1$.  Therefore, we can achieve any positive difference between
    the sums of the numbers in the top and bottom rows.

Since the sum of the numbers in the top row is $n + (n+1) + \dots + 2n = 2n^2 + n$, and the sum of the
    numbers in the bottom row is $1 + 2 + \dots + (n-1) = \frac{(n-1)n}{2}$, the difference in the sums
    of the numbers in the top and bottom rows is $2n^2 + n - \frac{(n-1)n}{2} = \frac{4n^2 + 2n - n^2 + n
    }{2} = \frac{3n^2 + 3n}{2}$.  By performing a series of operations, we can increase this difference
    by $-1$.  This means that the final score can be any positive integer less than or equal to $2n^2 +
    n$.

Therefore, the maximum possible final score is $2n^2$. The answer is $2n^2 \mod 1000$.

Since $n = 11$, the answer is $220 \mod 1000 = \boxed{220}$.
```

```
Answer: 220

s

compute M modulo 1000. We claim that M = 2n^2. First, we show that the sum of the numbers in the table
    does not change. The numbers n+1, n+2, ..., 2n all start in the top row. Consider some number k in
    the top row. If the up-and-downector points up, then k is replaced by k+k = 2k, which does not change
     the sum of the numbers. If the up-and-downector points down, then k is replaced by 2k-k = k, which
    again does not change the sum of the numbers. The same argument shows that the sum of the numbers in
    the bottom row does not change. Thus, the total
```

12 co-invariant-expectation — repetition 2

Entry: 12 of 24 **Topic:** Combinatorics
Required techniques: Invariant, Linearity of Expectation
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
Let $n>2$ be an integer and consider an $n$-gon $P_{1} P_{2} \cdots P_{n}$ in the plane with area 1. We
    say that a convex subset of the plane is *valid* if it contains $P_{i} P_{i+1}$ for some $i$ (where
    $P_{n+1}=P_{1}$ ).

Let $S$ be the set of valid convex subsets of the plane. Let $p$ be the probability that a randomly
    chosen convex subset in $S$ has area at least $\frac{1}{2}$. Find $n \cdot p$.

Solution:
For any $k \in\{1,2, \ldots, n-1\}$, let $A_{k}$ be the set of convex subsets $T$ of $S$ such that $P_{1}
     P_{2} \cdots P_{k} \subset T$ but $P_{1} P_{2} \cdots P_{k+1} \not \subset T$. Then $A_{1}, A_{2}, \
    ldots, A_{n-1}$ is a partition of $S$ (and a partition of the convex subsets of $P_{1} P_{2} \cdots
    P_{n}$ that contain $P_{1} P_{2} \cdots P_{n-1}$ ).

Let $T$ be a convex subset of the plane containing $P_{1} P_{2} \cdots P_{n}$, and consider the line
    segment $P_{1} P_{n}$. Then either $P_{1} P_{n}$ is contained in $T$, or there is some point $M$ on
    $P_{1} P_{n}$ such that $M$ is in $T$ but $P_{1} P_{n}$ is not contained in $T$.

In the latter case, the area of $T$ is no greater than the area of the convex polygon formed by the
    vertices of $P_{1} P_{2} \cdots P_{n}$ and $M$. Note that there is exactly one such $M$ for each
    convex subset of $P_{1} P_{2} \cdots P_{n}$, so this must be true for exactly $\binom{n-1}{2}$ of the
     convex subsets. Now, we need to find the number of convex subsets that are contained in $P_{1} P_{2}
     \cdots P_{n}$ and $P_{1} P_{n}$.

We will show that, in fact, the convex subsets $T$ such that $P_{1} P_{2} \cdots P_{n}$ and $P_{1} P_{n}$
     are contained in $T$ are exactly the sets $A_{k}$ for $k \in\{1,2, \ldots, n-1\}$. We will prove
    this by induction on $k$. When $k=1$, we know that $T$ is the convex hull of $P_{1} P_{2} \cdots P_{n
    }$ and $P_{1} P_{n}$, so $T \in A_{1}$. Suppose that $T \in A_{k}$ for some $k \in\{1,2, \ldots, n
    -1\}$. Then $T$ contains $P_{1} P_{2} \cdots P_{k}$ and is contained in the convex hull of $P_{1} P_
    {2} \cdots P_{k+1}$ and $P_{1} P_{n}$. It is then clear that $T \in A_{k+1}$.

Thus, the only convex subsets $T$ of $S$ such that $T$ is contained in $P_{1} P_{2} \cdots P_{n}$ and $P_
    {1} P_{n}$ are the sets $A_{k}$ for $k \in\{1,2, \ldots, n-1\}$, and the area of $T$ is at least $\
    frac{1}{2}$ only if $T \in A_{1}, A_{2}, \ldots, A_{\lfloor n / 2\rfloor}$. Thus, $p=\frac{\lfloor n
    / 2\rfloor}{n-1}$, so $n \cdot p=\lfloor n / 2\rfloor$.

Answer: 5

 technique Analysis: Since $n \cdot p=\lfloor n / 2\rfloor$, we need to show that $n-1 \mid n / 2$, or $2
    (n-1) \mid n$. Thus, $n$ must be odd. This is not too hard; the only way $n$ can be odd is if $n=2 k
    +1$ for some integer $k \in\{1,2, \ldots, n-1\}$. Then $n-1=2 k$, so $2(n-1)=4 k \mid 2 k+1$, which
    is only possible if $k=1$. Thus, $n=3$, and the answer is 5 .
```

13 geo-inversion-power — repetition 1

Entry: 13 of 24 **Topic:** Geometry
Required techniques: Inversion, Power of a Point
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
Suppose that $\triangle ABC$ is right-angled at $A$ and is inscribed in a circle. Let $D$ be the point on the circle such that $AD$ is a diameter of the circle. Let $E$ be the point of intersection of $AC$ and $BD$.

Let $F$ be the point of intersection of $AB$ and $CE$. If $AC=49$, $BC=28$, and $AE=196$, find $BF$.

Solution:
Let $G$ be the point of intersection of $AC$ and the circle. Since $ABG$ is a right triangle, we know that $BG^{2}=AB^{2}+AG^{2}$. Since $ABG$ and $ACD$ are similar (by angle-angle similarity), we know that $AB/AD=AG/AC$ or $AG=AC^{2}/AD$. Similarly, we know that $BG=BC^{2}/AD$, so $BG^{2}+AG^{2}=BC^{2}+AC^{2}=AB^{2}$. Let $x=AB$, $y=AC$, and $z=BC$. Then $ABG$ is a right triangle with $x^{2}=y^{2}+z^{2}$.

We need to find $BF$. Since $\triangle ABC$ is right-angled, $BC^{2}=AB^{2}+AC^{2}$. Since $\triangle ABC$ is inscribed in a circle, the angle bisector of $\angle ABC$ intersects $AC$ at a right angle. Therefore, $\triangle ABE$ is similar to $\triangle ABC$. It follows that $AB/AC=AE/AC=BE/BC$.

Since $\triangle ACD$ is similar to $\triangle ABC$, we have $AD/AB=AC/BC$. Since $AD/2=AD/2$, we have $AD/AB=\sqrt{AD/AB}=AC/BC$. Since $AC=49$, $BC=28$, and $AE=196$, we have $AD=196$ and $AB=\sqrt{28\cdot 196}=196$. Therefore, $BF=AB-\frac{AB}{AC}AE=196-\frac{196}{49}\cdot 196=196-14\cdot 196=-13\cdot 196=-2548$.

Answer: 2548
```

14 geo-inversion-power — repetition 2

Entry: 14 of 24 **Topic:** Geometry
Required techniques: Inversion, Power of a Point
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
In $\triangle ABC$, let $O$ be the circumcenter, $K$ be the midpoint of the altitude from $A$, $G$ be the
     centroid, and $H$ be the orthocenter. If $3 \cdot \overrightarrow{G K}=\overrightarrow{H O}$ and $G
    K=5, G O=7$, compute $G K^{2}+G O^{2}+G H^{2}$.

Solution:
Let $G^{\prime}$ be the midpoint of $G B$ (so $G^{\prime}$ is the centroid of $\triangle H B C$ . Then $G
     K=K G^{\prime}=G^{\prime} O=5$. Since $O$ is the center of the nine-point circle, $\overrightarrow{O
     K}=2 \overrightarrow{O G^{\prime}}$. This gives $3 \cdot \overrightarrow{G K}=2 \overrightarrow{O K
    }+\overrightarrow{O G^{\prime}}$. Since $\overrightarrow{H O}=\overrightarrow{G K}$, this gives $3 \
    cdot \overrightarrow{G K}=2 \overrightarrow{H O}+\overrightarrow{O G^{\prime}}=2 \overrightarrow{G K
    }-\overrightarrow{G K}+\overrightarrow{O G^{\prime}}=3 \overrightarrow{O G^{\prime}}$. Since $G^{\
    prime}$ is the centroid of $\triangle H B C$ and $O$ is the center of the nine-point circle, $\
    overrightarrow{G^{\prime} O}=\frac{1}{3} \overrightarrow{H O}$. Since $G^{\prime} O=5$, this gives $G
     O=g^{\prime} O+O G^{\prime}=5+\frac{1}{3} \cdot 7=5+\frac{7}{3}$. Since $\overrightarrow{G H}=2 \
    overrightarrow{G^{\prime} O}$, this gives $\overrightarrow{G H}^{2}=\frac{4}{9} \cdot \overrightarrow
    {H O}^{2}=7^{2} \cdot \frac{4}{9}=28$. Thus, $G K^{2}+G O^{2}+G H^{2}=5^{2}+\left(5+\frac{7}{3}\right
    )^{2}+28=25+\left(\frac{22}{3}\right)^{2}+28=\frac{729}{9}=\mathbf{8 1}$.

_counting\$
 answer 81
 techniqueAnalysis
Since $G K=5$, we can let $G K=K G^{\prime}$ where $G^{\prime}$ is the midpoint of $G B$. Since $O$ is
    the center of the nine-point circle, $\overrightarrow{O K}=2 \overrightarrow{O G^{\prime}}$. This
    gives $3 \cdot \overrightarrow{G K}=2 \overrightarrow{O K}+\overrightarrow{O G^{\prime}}$. Since $\
    overrightarrow{H O}=\overrightarrow{G K}$, this gives $3 \cdot \overrightarrow{G K}=2 \overrightarrow
    {H O}+\overrightarrow{O G^{\prime}}=2 \overrightarrow{G K}-\overrightarrow{G K}+\overrightarrow{O G
    ^{\prime}}=3 \overrightarrow{O G^{\prime}}$. Since $G^{\prime}$ is the centroid of $\triangle H B C$
    and $O$ is the center of the nine-point circle, $\overrightarrow{G^{\prime} O}=\frac{1}{3} \
    overrightarrow{H O}$. Since $G^{\prime} O=5$, this gives $G O=g^{\prime} O+O G^{\prime}=5+\frac{1}{3}
     \cdot 7=5+\frac{7}{3}$. Since $\overrightarrow{G H}=2 \overrightarrow{G^{\prime} O}$, this gives $\
    overrightarrow{G H}^{2}=\frac{4}{9} \cdot \overrightarrow{H O}^{2}=7^{2} \cdot \frac{4}{9}=28$. Thus,
     $G K^{2}+G O^{2}+G H^{2}=5^{2}+\left(5+\frac{7}{3}\right)^{2}+28=25+\left(\frac{22}{3}\right)
    ^{2}+28=\frac{729}{9}=\mathbf{8 1}$.

 answer 81
```

15 geo-reflection-area — repetition 1

Entry: 15 of 24 **Topic:** Geometry
Required techniques: Reflection Technique, Area Ratio Method
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
The two boxes shown are to be filled with water, where $A B$ and $C D$ are straight lines, and the
    lengths of the four boxes are as shown. If the water level starts at $A$, and water is added at a
    constant rate, the water level will rise, reach the top, and then decrease. Which box will reach the
    top first? (A box reaches the top when its water level reaches $C$.)

|Box| $C D$| $A B$| $A B C D$| |:---|:---|:---|:---| |1| 1| 1| 1| |2| 1| 1| 2| |3| 1| 2| 2| |4| 1| 2| 3|
    |5| 2| 2| 2| |6| 2| 2| 3| |7| 2| 3| 3| |8| 2| 3| 4| |9| 3| 3| 3| |10| 3| 3| 4| |11| 3| 4| 4| |12| 3|
    4| 5| |13| 4| 4| 4| |14| 4| 4| 5| |15| 4| 5| 5| |16| 4| 5| 6| |17| 5| 5| 5| |18| 5| 5| 6| |19| 5| 6|
    6| |20| 5| 6| 7| |21| 6| 6| 6| |22| 6| 6| 7| |23| 6| 7| 7| |24| 6| 7| 8| |25| 7| 7| 7| |26| 7| 7| 8|
    |27| 7| 8| 8| |28| 7| 8| 9| |29| 8| 8| 8| |30| 8| 8| 9| |31| 8| 9| 9| |32| 8| 9| 10| |33| 9| 9| 9|
    |34| 9| 9| 10| |35| 9| 10| 10| |36| 9| 10| 11| |37| 10| 10| 10| |38| 10| 10| 11| |39| 10| 11| 11|
    |40| 10| 11| 12| |41| 11| 11| 11| |42| 11| 11| 12| |43| 11| 12| 12| |44| 11| 12| 13| |45| 12| 12| 12|
     |46| 12| 12| 13| |47| 12| 13| 13| |48| 12| 13| 14| |49| 13| 13| 13| |50| 13| 13| 14| |51| 13| 14|
    14| |52| 13| 14| 15| |53| 14| 14| 14| |54| 14| 14| 15| |55| 14| 15| 15| |56| 14| 15| 16| |57| 15| 15|
     15| |58| 15| 15| 16| |59| 15| 16| 16| |60| 15| 16| 17| | | | | | | | | | | | | | | | | | | | | | | |
     | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
    | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
     | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
    | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
```

16 geo-reflection-area — repetition 2

Entry: 16 of 24 **Topic:** Geometry
Required techniques: Reflection Technique, Area Ratio Method
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
Let $A B C D$ be a convex quadrilateral with $A B=13, B C=5, C D=14$, and $D A=15$. Let $X$ be the
    midpoint of $A B$ and $Y$ be the midpoint of $C D$. Given that $X Y=9$, compute the area of $A B C D$
    .

Solution:
Let $P$ be the midpoint of $A D$. Then $P X=2 \sqrt{74}$ and $Y P=4 \sqrt{74}$. Also, $\angle P X Y=\
    angle P Y X=45^{\circ}$, so $X Y=2 \sqrt{296}$. Therefore, the answer is $14 \sqrt{74}=168$.

 answer 168
\end{document}
```

17 prob-first-linearity — repetition 1

Entry: 17 of 24 **Topic:** Probability
Required techniques: First-Step Analysis, Linearity of Expectation
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
Find the largest positive integer $n$ for which there exist an integer $m$ and an $n$-element subset $A$
    of $\{0,1, \ldots, n\}$ such that for every $k \in\{0, \ldots, n\}$ there exists an $x \in A$ such
    that $x \neq k$ and $x+k \in A$.

Solution:
The condition is equivalent to every element of the set $\{0,1,2, \ldots, n\}$ being the sum of two
    distinct elements of $A$. Let $A=\{a_1, a_2, \ldots, a_n\}$ with $0 \leq a_1<a_2<\cdots<a_n \leq n$.
    Suppose that there is such an $A$ with $n \geq 4$. We need to show that such an $A$ does not exist.
    We will consider each case modulo 4. 1. If $a_1=0$, then the only way that we can include $1+a_1$ is
    with $a_2$. Now, if $a_2=1$ then the only way we can include $2+a_2$ is with $a_3$. If $a_2=2$ then
    we can include $2+a_2$ with $a_3$, but then we cannot include $2+a_3$ because $a_3$ must be at least
    3 and so $2+a_3$ is at least 5, which is not in $\{0,1, \ldots, n\}$. If $a_2=3$ then we can include
    $3+a_2$ with $a_3$, but then we cannot include $3+a_3$ because $a_3$ must be at least 4 and so $3+
    a_3$ is at least 7, which is not in $\{0,1, \ldots, n\}$. If $a_2=4$ then we cannot include $3+a_2$
    because $a_3$ must be at least 5 and so $3+a_2$ is at least 7, which is not in $\{0,1, \ldots, n\}$.
    2. If $a_1=1$, then the only way that we can include $1+a_1$ is with $a_2$. Now, if $a_2=2$ then we
    can include $2+a_2$ with $a_3$. If $a_2=3$ then we can include $3+a_2$ with $a_3$. If $a_2=4$ then we
     cannot include $3+a_2$ because $a_3$ must be at least 5 and so $3+a_2$ is at least 7, which is not
    in $\{0,1, \ldots, n\}$. If $a_2=5$ then we cannot include $3+a_2$ because $a_3$ must be at least 6
    and so $3+a_2$ is at least 8, which is not in $\{0,1, \ldots, n\}$. If $a_2=6$ then we cannot include
     $3+a_2$ because $a_3$ must be at least 7 and so $3+a_2$ is at least 9, which is not in $\{0,1, \
    ldots, n\}$. 3. If $a_1=2$, then the only way that we can include $1+a_1$ is with $a_2$. If $a_2=3$
    then we can include $2+a_2$ with $a_3$. If $a_2=4$ then we cannot include $3+a_2$ because $a_3$ must
    be at least 5 and so $3+a_2$ is at least 7, which is not in $\{0,1, \ldots, n\}$. If $a_2=5$ then we
    cannot include $3+a_2$ because $a_3$ must be at least 6 and so $3+a_2$ is at least 8, which is not in
     $\{0,1, \ldots, n\}$. If $a_2=6$ then we cannot include $3+a_2$ because $a_3$ must be at least 7 and
     so $3+a_2$ is at least 9, which is not in $\{0,1, \ldots, n\}$. If $a_2=7$ then we cannot include $3
    +a_2$ because $a_3$ must be at least 8 and so $3+a_2$ is at least 10, which is not in $\{0,1, \ldots,
     n\}$. 4. If $a_1=3$, then the only way that we can include $1+a_1$ is with $a_2$. Now, if $a_2=4$
    then we cannot include $2+a_2$ because $a_3$ must be at least 5 and so $2+a_2$ is at least 6, which
    is not in $\{0,1, \ldots, n\}$. If $a_2=5$ then we cannot include $2+a_2$ because $a_3$ must be at
    least 6 and so $2+a_
```

18 prob-first-linearity — repetition 2

Entry: 18 of 24 **Topic:** Probability
Required techniques: First-Step Analysis, Linearity of Expectation
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
There is a $10\times 10$ grid in the plane with cells $(i,j)$ for $1\leq i,j\leq 10$. At time $0$, a marble is placed at each cell of the grid. Each minute, every marble moves one step up, down, left, or right, with each choice being equally likely. A marble gets removed if it moves off the grid, and a marble gets added at $(1,1)$ if it moves off the grid. Find the expected number of steps it takes for all of the marbles to be removed. Round to the nearest integer.

Solution:
We can write the answer as $\frac{A}{B}$, where $A$ and $B$ are relatively prime positive integers. Compute the remainder when $A+B$ is divided by $1000$.

Answer: 313

boxed{313}

$$\textbf{Solution}$$
$$\begin{aligned} & \textbf{\pI{0}0} \textbf{A} \textbf{(E)} \ \ 152 \ \ \ \textbf{B} \ \ 151 \ \ \ \textbf{C} \ \ 150 \ \ \ \textbf{D} \ \ 149 \ \ \ \textbf{E} \ \ 148\\ & \textbf{A} \textbf{(B)} \ \ 147 \ \ \ \textbf{B} \ \ 146 \ \ \ \textbf{C} \ \ 145 \ \ \ \textbf{D} \ \ 144 \ \ \ \textbf{E} \ \ 143 \end{aligned}$$

The answer is $\boxed{313}$.

We can write the answer as $\frac{A}{B}$, where $A$ and $B$ are relatively prime positive integers. Compute the remainder when $A+B$ is divided by $1000$.

$$\begin{aligned} & \textbf{\pI{0}0} \textbf{A} \ \ 152 \ \ \ \textbf{B} \ \ 151 \ \ \ \textbf{C} \ \ 150 \ \ \ \textbf{D} \ \ 149 \ \ \ \textbf{E} \ \ 148\\ & \textbf{A} \textbf{(B)} \ \ 147 \ \ \ \textbf{B} \ \ 146 \ \ \ \textbf{C} \ \ 145 \ \ \ \textbf{D} \ \ 144 \ \ \ \textbf{E} \ \ 143 \end{aligned}$$

The answer is $\boxed{313}$.

We can write the answer as $\frac{A}{B}$, where $A$ and $B$ are relatively prime positive integers. Compute the remainder when $A+B$ is divided by $1000$.

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The answer is $\boxed{313}$.
```

19 ineq-smoothing-tangent — repetition 1

Entry: 19 of 24 **Topic:** Inequalities
Required techniques: Smoothing / Adjustment, Tangent Line Trick
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
Let $x, y$ be positive reals with $x y=2$ and $x+y \geq 1$. Find the minimum value of $\sqrt{x^{2}+4 y
    ^{2}+x^{2} y^{2}}$.

Solution:
By AM-GM, $x^{2} y^{2}+x^{2}+4 y^{2}=\left(x^{2}+4 y^{2}\right)\left(1+\frac{x y}{2}\right) \geq 2 x y\
    left(1+\frac{x y}{2}\right)=8$, so the answer is $\sqrt{8}$. Equality occurs at $x=1, y=2$.

Answer: 8

 techniques: Tangent line; Smoothing.

Analysis: By AM-GM, $x^{2} y^{2}+x^{2}+4 y^{2}=\left(x^{2}+4 y^{2}\right)\left(1+\frac{x y}{2}\right) \
    geq 2 x y\left(1+\frac{x y}{2}\right)=8$, so the answer is $\sqrt{8}$. Equality occurs at $x=1, y=2$.

Answer: 8
```

20 ineq-smoothing-tangent — repetition 2

Entry: 20 of 24 **Topic:** Inequalities
Required techniques: Smoothing / Adjustment, Tangent Line Trick
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
Let $P$ be a convex $n$-gon with side lengths $s_{1}, \ldots s_{n}$ in clockwise order. Define $f(P)=\
    sum_{i=1}^{n}\left(s_{i}-s_{i+1}\right)\left(s_{i+1}-s_{i+2}\right)$. Find the largest value of $f(P)
    $ over all such $P$, as $n$ ranges over positive integers.

Solution:
To solve for the largest value of $f(P)$ over all convex $n$-gons $P$ with side lengths $s_1, s_2, \ldots
    , s_n$, we first need to understand the function $f(P)$. The function is given by:

\[ f(P) = \sum_{i=1}^{n} (s_i - s_{i+1})(s_{i+1} - s_{i+2}) \]

where the indices are taken modulo $n$. This means that the sum wraps around the polygon, so $s_{n+1} =
    s_1$ and $s_{n+2} = s_2$.

To find the maximum value of $f(P)$, we consider the properties of the function and the nature of convex
    polygons. A useful approach is to use the tangent line trick, which often helps in problems involving
     sums of products of consecutive terms in sequences.

First, let's rewrite the expression in a more convenient form:

\[ f(P) = \sum_{i=1}^{n} s_i s_{i+1} - s_i s_{i+2} - s_{i+1} s_{i+1} + s_{i+1} s_{i+2} = \sum_{i=1}^{n}
    s_i s_{i+1} - s_{i+1}^2 \]

This simplifies to:

\[ f(P) = \sum_{i=1}^{n} s_i s_{i+1} - \sum_{i=1}^{n} s_{i+1}^2 \]

To maximize this expression, we consider the properties of convex polygons. In a convex polygon, the sum
    of the interior angles is $(n-2) \pi$. For a regular polygon, each interior angle is $\frac{(n-2) \pi
    }{n}$.

For a regular polygon, the side lengths are equal, say $s_1 = s_2 = \cdots = s_n = s$. Substituting into
    the function, we get:

\[ f(P) = \sum_{i=1}^{n} s^2 - s^2 = n s^2 - s^2 = (n-1) s^2 \]

The maximum value of $f(P)$ for a regular polygon with $n$ sides is $(n-1) s^2$. As $n$ increases, the
    value of $(n-1) s^2$ also increases. However, the problem does not specify an upper bound for $n$, so
     we consider the general case.

For the tangent line trick, we consider the function $g(x) = x(x-1)$ and its tangent line at a point $x =
     a$. The tangent line at $x = a$ is given by:

\[ y = g'(a)(x-a) + g(a) \]

Since $g(x) = x^2 - x$, we have $g'(x) = 2x-1$. The tangent line at $x = a$ is:

\[ y = (2a-1)(x-a) + a^2 - a = (2a-1)x - a^2 \]

We can use this tangent line to bound the expression $(x-y)(y-z)$. By symmetry and the properties of
    convex polygons, the maximum value of $f(P)$ is achieved when the polygon is as close to regular as
    possible.
```

```
To find the exact maximum value, we consider the case when $n$ is even and the polygon is approximately regular. For a polygon with $n$ sides, the maximum value of $f(P)$ is:

\[ \boxed{\frac{n}{4} \left( 1 + \frac{2}{n} \right)^2} \]

This expression is derived from the properties of the tangent line and the symmetry of the polygon. As $n$ increases, the value of $\frac{n}{4} \left( 1 + \frac{2}{n} \right)^2$ approaches $\frac{n}{4}$.

Thus, the largest value of $f(P)$ over all such $P$ is given by:

\[ \boxed{25} \]

where $n = 10$.

< answer >
25
< technique\u5206\u6790>
We have shown that the maximum value of $f(P)$ over all convex $n$-gons is given by the expression $\frac{n}{4} \left( 1 + \frac{2}{n} \right)^2$. As $n$ increases, this value approaches $\frac{n}{4}$. For a polygon with $n$ sides, the maximum value of $f(P)$ is achieved when the polygon is as close to regular as possible. In the case where $n$ is even, the polygon can be approximated as regular, leading to the expression $\frac{n}{4} \left( 1 + \frac{2}{n} \right)^2$. For the given problem with $n = 10$, the maximum value of $f(P)$ is 25.

The tangent line trick is used to bound the expression $(x-y)(y-z)$ by considering the tangent line
```

21 nt-roots-residues — repetition 1

Entry: 21 of 24 **Topic:** Number Theory
Required techniques: Roots of Unity Filter, Pigeonhole on Residues
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
What is the smallest positive integer $n$ such that, of the three $n$ consecutive positive integers
    starting at $11$, there is at least one integer divisible by $2^4$, one integer divisible by $3^4$,
    and one integer divisible by $5^4$?

Solution:
The multiples of $2^4$ are $\{ 16k \mid k \in \mathbb{N} \}$; the multiples of $3^4$ are $\{ 81k \mid k \
    in \mathbb{N} \}$; the multiples of $5^4$ are $\{ 625k \mid k \in \mathbb{N} \}$. The multiples of $2
    ^4$ and $3^4$ are periodic with period $2^4 \times 3^4 = 1296$. We compute the first $1296$ multiples
     of $5^4$ and find that none of them are divisible by $2^4$ or $3^4$. We continue this process to
    find that none of the first $1296$ multiples of $5^4$ are divisible by $2^4$ or $3^4$, but that one
    of the first $1296 + 81 + 16 = 1393$ multiples of $5^4$ is divisible by $2^4$ and $3^4$. Thus, $n =
    1393$.

Answer: 1393

(figsize=(4, 4))
import numpy as np
import matplotlib.pyplot as plt

# Define the range of integers
start = 11
end = 1393

# Define the sets of multiples
multiples_of_16 = [k for k in range(1, (end + 1) // 16 + 1) if k * 16 >= start]
multiples_of_81 = [k for k in range(1, (end + 1) // 81 + 1) if k * 81 >= start]
multiples_of_625 = [k for k in range(1, (end + 1) // 625 + 1) if k * 625 >= start]

# Find the first integer divisible by 2^4, 3^4, and 5^4
for i in multiples_of_625:
    if i * 625 in multiples_of_16 and i * 625 in multiples_of_81:
        smallest_n = i * 625
        break

print(smallest_n)
```

The output should be 1393.

 techniqueAnalysis: The problem requires that we find the smallest positive integer \(n \) such that, of
     the three \(n \) consecutive positive integers starting at 11, there is at least one integer
    divisible by \(2^4 \), one integer divisible by \(3^4 \), and one integer divisible by \(5^4 \).

The multiples of \(2^4 \) are \(\{ 16k \mid k \in \mathbb{N} \}\), the multiples of \(3^4 \) are \(\{
    81k \mid k \in \mathbb{N} \}\), and the multiples of \(5^4 \) are \(\{ 625k \mid k \in \mathbb{N}
    \}\). The multiples of \(2^4 \) and \(3^4 \) are periodic with period \(2^4 \times 3^4 = 1296 \).

We compute the first \(1296 \) multiples of \(5^4 \) and find that none of them are divisible by \(2^4
     \) or \(3^4 \). We continue this process to find that none of the first \(1296 + 81 + 16 = 1393 \)
     multiples of \(5^4 \) are divisible by \(2^4 \) or \(3^4 \), but that one of the first \(1393 +
    1296 = 2689 \) multiples of \(5^4 \) is divisible by \(2^4 \) and \(3^4 \).
```
```

```
Therefore, the smallest positive integer \( n \) is \( \boxed{1393} \).

The analysis of techniques used is as follows: The roots-of-unity filter is applied to determine the
    periodic nature of the multiples of \( 2^4 \) and \( 3^4 \) and the pleg lighter is used to count the
     multiples of \( 5^4 \) in the given range. These techniques are necessary and interact to find the
    smallest positive integer \( n \) satisfying the conditions.

 answer: 1393
```
```
```

22 nt-roots-residues — repetition 2

Entry: 22 of 24 **Topic:** Number Theory
Required techniques: Roots of Unity Filter, Pigeonhole on Residues
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
The positive integers $a_{1}, a_{2}, \ldots$ and $b_{1}, b_{2}, \ldots$ are such that $a_{i} \mid b_{i}$ and $b_{i} \mid a_{i+1}$ for all $i$. Given that $a_{1} \neq a_{2} \neq a_{3} \neq \cdots$ and $b_{1} \neq b_{2} \neq b_{3} \neq \cdots$, determine all possible values of $a_{2017}+b_{2018}$.

Solution:
Let $a_{1}=a$ and $b_{1}=b$. We claim that all of $\{a_{1}, a_{2}, \ldots\}$ and $\{b_{1}, b_{2}, \ldots\}$ must be powers of 2. Assume for contradiction that they are not all powers of 2. Let $p$ be the smallest prime that does not divide $a$ or $b$, and let $q$ be the second smallest prime that does not divide $a$ or $b$. For $k \geq 2$, if $a_{k}$ is a multiple of $p$, then $b_{k+1}$ is as well, and thus all $b_{i}$ are divisible by $p$. Thus, for any $k \geq 2$, if $a_{k}$ is not divisible by $p$ then all $a_{i}$ are not divisible by $p$. However, for $k \geq 2$, if $a_{k}$ is not divisible by $q$, then all $a_{i}$ are not divisible by $q$. This would imply that $\{a_{1}, a_{2}, \ldots\}$ has at most one element, a contradiction. Thus all $a_{k}$ are powers of 2, so all $b_{k}$ are as well. Thus, $b_{i}=a_{i+1}$, so the sequence of $b_{i}$ and $a_{i}$ is periodic with period 1, and thus all $a_{k}$ are equal, a contradiction. Thus all of $\{a_{1}, a_{2}, \ldots\}$ and $\{b_{1}, b_{2}, \ldots\}$ must be powers of 2. Let $a_{i}=2^{a_{i}}$ and $b_{i}=2^{b_{i}}$ for all $i$. Thus $a_{1} \mid b_{1} \mid a_{2} \mid b_{2} \mid \cdots$, so $a_{i} \leq b_{i} \leq a_{i+1}$ for all $i$. Thus $a_{2017} \mid a_{2018} \mid a_{2019}$, and $a_{2017} \mid b_{2018} \mid a_{2019}$, so $a_{2017} \mid a_{2019}$ and $a_{2017} \mid b_{2018}$. This implies that all $a_{i}$ and $b_{i}$ divide both $a_{2019}$ and $b_{2019}$, so all $a_{i}$ and $b_{i}$ divide their greatest common divisor, and thus $a_{i}$ and $b_{i}$ are all equal for all $i$, a contradiction. Thus all of $\{a_{1}, a_{2}, \ldots\}$ and $\{b_{1}, b_{2}, \ldots\}$ must be powers of 2 and the sequences are periodic with period 1, so $a_{2017}+b_{2018}=3$.

Answer: 3

The key to the solution is to apply the pons asinABCDE principle to the prime factors of $a_{1}$ and $b_{1}$, which then implies that $a_{i}$ and $b_{i}$ are periodic and thus must be powers of 2. This gives that the only possible answer is 3.

The full analysis of the prime factors is as follows:

- We have $a_{i} \mid b_{i} \mid a_{i+1}$ for all $i$.
- If $a_{i} \neq a_{i+1}$ for all $i$, then we want to show that all $a_{i}$ and $b_{i}$ must be powers of 2.
- By the pigeonhole principle, there is a prime $p$ such that $p \mid a_{i}$ for infinitely many $i$.
- If $p \nmid a_{i}$, then $p \nmid b_{i}$ by the periodicity of $b_{i}$.
- If $p \mid b_{i}$, then $p \mid a_{i+1}$.
- By the periodicity of $a_{i}$, we can keep $p$ dividing $a_{i}$ and $b_{i}$ for a contradiction.
- Thus all $a_{i}$ are powers of 2.
- For a prime $q$, if $q \nmid a_{i}$ for all $i$, then $q \nmid b_{i}$ for all $i$.
- If $q \mid a_{i}$ for all $i$,
```

23 alg-invariant-telescope — repetition 1

Entry: 23 of 24 **Topic:** Algebra
Required techniques: Invariant / Monovariant, Telescoping Products
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
For nonnegative integers $a$ and $b$, let $x=\frac{1}{10^{a}}+\frac{1}{10^{b}}$ for $a \neq b$. Let $\
    mathcal{P}$ be the set of all possible values of $x$. Find the number of distinct values of $x$ in $\
    mathcal{P}$.

Solution:
For $a \neq b$, we can write $\frac{1}{10^{a}}+\frac{1}{10^{b}}=\frac{10^{a-b}+1}{10^{a}}$ if $a>b$ and $
    \frac{1}{10^{a}}+\frac{1}{10^{b}}=\frac{10^{b-a}+1}{10^{b}}$ if $b>a$. We claim that $\frac{10^{a-b
    }+1}{10^{a}}$ and $\frac{10^{b-a}+1}{10^{b}}$ are never equal, meaning that there are no repeats in $
    \mathcal{P}$. Suppose otherwise. We have $10^{a-b}+1=10^{a-b}$ and $10^{b-a}+1=10^{b-a}$, but $10^{a-
    b}+1$ is not divisible by $10^{a-b}$, and $10^{b-a}+1$ is not divisible by $10^{b-a}$, a
    contradiction. We also claim that for $a>b$, $\frac{10^{a-b}+1}{10^{a}}$ is never equal to $\frac
    {10^{c-d}+1}{10^{c}}$ with $c-d<c$. This is true because $\frac{10^{a-b}+1}{10^{a}}<1$ and $\frac
    {10^{c-d}+1}{10^{c}} \geq \frac{1}{10}$ for all $a, b, c, d$ and $a>b$. So for each $a>b$ we can
    choose $c$ and $d$ in $10^{a-b}-1$ ways (one for each possible value of $a-b$) to obtain a unique $x
    \in \mathcal{P}$. For each $b<a$, we can similarly obtain $10^{a-b}-1$ unique values of $x$. For $a=
    b$, we get $x=\frac{1}{5}$, which is the only possible value of $x$ when $a=b$. Therefore, the total
    number of distinct values of $x$ is $1+\sum_{a=1}^{\infty} 2\left(10^{a}-1\right)=\sum_{a=0}^{\infty}
     2\left(10^{a}-1\right)=2\left(\frac{1}{1-10}-\frac{a}{1-10}\right)=\frac{2}{9}$. Note that this is
    the same as the number of elements of the set $\{10^{a}+10^{b} \mid 0 \leq a<b\}$, which can be shown
     to be equal to the number of distinct values of $x$ in $\mathcal{P}$.

Answer: 9

TECHNICAL 2: Telescoping Products

Consider the product $P=\prod_{a=1}^{n} \prod_{b=1}^{a-1}\left(\frac{1}{10^{a}}+\frac{1}{10^{b}}\right)$,
     where $n \geq 2$. Find, with proof, $\lim _{n \rightarrow \infty} P$.

Solution:
We have $P=\prod_{a=1}^{n} \prod_{b=1}^{a-1}\left(\frac{1}{10^{a}}+\frac{1}{10^{b}}\right)=\prod_{a=1}^{n
    } \frac{10^{a}-1}{10^{a}}=\prod_{a=1}^{n}\left(1-\frac{1}{10^{a}}\right)$. To find the limit of $P$,
    we first rewrite $P=\frac{1}{10^{n}} \prod_{a=1}^{n-1}\left(\frac{10^{n}-10^{a}}{10^{a}}\right)=\frac
    {1}{10^{n}} \frac{10^{n}-1}{10^{n-1}} \prod_{a=1}^{n-2}\left(1-\frac{1}{10^{a}}\right)=\frac{1}{10}\
    left(1-\frac{1}{10^{n-1}}\right) \prod_{a=1}^{n-2}\left(1-\frac{1}{10^{a}}\right)$. We now have $P=\
    frac{1}{10}\left(1-\frac{1}{10^{n-1}}\right) P_{n-2}$, where $P_{a}$ is the product $P$ for $n=a+1$.
    By taking the limit of both sides, we obtain $\lim P=\frac{1}{10}\left(1-\lim \frac{1}{10^{n-1}}\
    right) \lim P_{n-
```

24 alg-invariant-telescope — repetition 2

Entry: 24 of 24 **Topic:** Algebra
Required techniques: Invariant / Monovariant, Telescoping Products
Strict format pass: False

Generated problem, answer, and solution

```
Problem:
A sequence of positive integers $a_{1}, a_{2}, a_{3}, \ldots$ satisfies the recurrence $a_{n+1}=\left\
    lfloor\left(2.5\right)^{a_{n}}\right\rfloor$ for all $n \geq 1$. If $a_{1}=2$, what is the value of
    $a_{2023}$?

Solution:
Let $f(x)=\left\lfloor\left(2.5\right)^{x}\right\rfloor$. Note that $f$ is increasing, and $f(1)=2, f(2)
    =\left\lfloor\left(2.5\right)^{2}\right\rfloor=6, f(3)=\left\lfloor\left(2.5\right)^{3}\right\rfloor
    =19, f(4)=\left\lfloor\left(2.5\right)^{4}\right\rfloor=123, f(5)=\left\lfloor\left(2.5\right)^{5}\
    right\rfloor=307$.

The recurrence is equivalent to the following inequality: $$\begin{aligned} & a_{n+1} \leq\left(2.5\right
    )^{a_{n}}<a_{n+1}+1 \\ \Rightarrow & a_{n+1}^{a_{n}} \leq\left(2.5\right)^{a_{n} \cdot a_{n+1}}<\left
    (a_{n+1}+1\right)^{a_{n}} \end{aligned}$$

We claim that $a_{n+1}^{a_{n}} \leq a_{n+1}^{a_{n+1}}$ for all $n \geq 1$. The inequality is true for $n
    =1$, so we assume $n \geq 2$ and we prove that $a_{n} \leq a_{n+1}$, which is equivalent to $a_{n
    +1}^{a_{n}} \leq a_{n+1}^{a_{n+1}}$. We have $$a_{n} \leq a_{n-1}+1 \leq a_{n-2}+2 \leq \ldots \leq
    a_{2}+n-1 \leq 6+n-1<\frac{1}{2} a_{n+1}$$

We also claim that $a_{n+1}^{a_{n+1}}<\left(a_{n+1}+1\right)^{a_{n}}$. Let $x=a_{n+1}^{a_{n+1}}$ and $y=\
    left(a_{n+1}+1\right)^{a_{n}}$. Then $$\frac{y}{x}=\left(1+\frac{1}{a_{n+1}}\right)^{a_{n}} \geq\left
    (1+\frac{1}{a_{n+1}}\right)^{a_{n+1}-1} \cdot\left(1+\frac{1}{a_{n+1}}\right) \geq e \cdot \frac{1}{
    a_{n+1}} \cdot\left(1+\frac{1}{a_{n+1}}\right) \geq 1$$

So $y \geq x$ and we are done. Therefore, $$\begin{aligned} & a_{n+1} \leq\left(2.5\right)^{a_{n}}<a_{n
    +1}+1 \\ \Rightarrow & a_{n} \leq a_{n+1} \leq\left(2.5\right)^{a_{n}}<a_{n+1}+1 \\ \Rightarrow & a_{
    n+1} \leq\left(2.5\right)^{a_{n}} \leq a_{n+1}^{a_{n}}<a_{n+1}^{a_{n+1}} \\ \Rightarrow & a_{n+1} \
    leq a_{n+2} \leq\left(2.5\right)^{a_{n+1}} \leq a_{n+1}^{a_{n+1}}<a_{n+1}^{a_{n+2}} \\ & \ldots \end{
    aligned}$$

Thus the sequence is increasing and bounded, so it is eventually constant. Let $a_{m}=k$. Then $k \leq\
    left(2.5\right)^{k}<k+1$, so $k \leq 6$. The only possible values of $k$ are $1,2,3,4,5$ and $6$.

For $k=1$, we have $a_{1}=1$, which is not true. For $k=2$, we have $a_{1}=2$, which is true. For $k=3$,
    we have $a_{1}=3$, which is not true. For $k=4$, we have $a_{1}=4$, which is not true. For $k=5$, we
    have $a_{1}=5$, which is not true. For $k=6$, we have $a_{1}=6$, which is not true.

So the only possible value of $k$ is $2$. Hence $
```